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Maternal substance use disorder and neonatal health: epidemiologic and socioeconomic determinants in a case-control study, Shiraz, Iran

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Abstract

Background Neonates born to mothers with substance use disorder may require pharmacological or non-pharmacological treatments for withdrawal symptoms post-birth. This study aims to assess parents' epidemiologic and socioeconomic factors and their correlation with short neonatal outcomes, such as length of hospital stay and discharge disposition.

Methods This case-control study was conducted in hospitals affiliated with Shiraz University of Medical Sciences. Demographic and social data of parents and neonatal outcomes, including hospital stay, Finnegan score, and discharge disposition, were extracted from hospital records. Statistical analysis was performed using SPSS, reporting means, percentages, and frequencies. Independent sample t-tests, ANOVA, and chi-square tests were used to evaluate associations between mothers' socio-demographic background and clinical status. Negative binomial regression was employed to explore risk factors for length of hospital stay.

Results The case group comprised 95 neonates born to mothers with substance use disorder, with 122 healthy neonates as the control group. The incidence of birth from mothers with substance use disorder was 1.4%. The mean maternal age was 32.1 ± 6.5 years. The average Finnegan score among affected neonates was 6.85 ± 1.85 . Methadone was the most commonly used substance. While all control group neonates were discharged home, only 57.9% of cases were discharged home. Discharge to social services was more common among unmarried and homeless or tenant mothers. Neonates born to mothers with academic degrees had 1.54 times longer hospital stay.

Conclusion This study underscores the intricate relationship between maternal substance abuse, socioeconomic factors, and neonatal health, highlighting the necessity for targeted interventions to support mothers with substance use disorder and their infants.

Keywords Neonatal abstinence syndrome, Substance-related disorders, Newborn, Infant, Socioeconomic factors, Maternal behavior

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Introduction

Neonatal abstinence syndrome (NAS), recently known as neonatal opioid withdrawal syndrome (NOWS), occurs when newborns experience withdrawal symptoms after being exposed to opioids or other drugs in utero [1]. The incidence of NOWS varies across regions, with an increasing trend observed [2, 3]. The incidence is reported at 7 in 1000 hospitalized births in the US [4]. In Iran, the prevalence of substance use during pregnancy has been reported to range from less than 5 to nearly 50 per 1,000 births, with regional variation depending on socioeconomic and cultural factors [5–10]. Opioid overdose occurrences are more prevalent among individuals possessing public insurance, residing in economically disadvantaged regions, and experiencing unemployment [11, 12].

The psychosocial impact of substance use during pregnancy includes stress, financial and legal issues, diminished social support, and partner violence. These factors can either lead to substance use or result from it [13, 14]. In a review article on the epidemiology and effects of substance use during pregnancy, several risk factors for substance use disorders during pregnancy were identified, including age below 25 years, high school education or less, income below the poverty level, psychiatric disorders, and a family history of substance use disorder [13]. Vesoulis et al. evaluated these social factors and found that poverty is associated with increased length of hospitalization [15]. Additionally, Singleton and colleagues demonstrated that maternal drug therapy for opioid misuse leads to improved hospitalization outcomes and an increased likelihood of discharge to home compared to childcare facilities [16].

Maternal substance abuse is the primary psychosocial risk factor prompting referrals to child protective services [17, 18].

Shiraz, as one of the largest cities in southern Iran and a referral hub for high-risk pregnancies and neonatal intensive care, has been identified as an area with a considerable burden of substance use disorders. Hospitals affiliated with Shiraz University of Medical Sciences admit neonates exposed to opioids and other substances, making this setting particularly suitable for investigating the epidemiologic, socioeconomic, and clinical determinants of neonatal outcomes in this vulnerable population.

These findings underscore the importance of addressing epidemiologic and social health determinants in the NOWS context. Despite the growing body of literature on neonatal abstinence syndrome, both globally and in Iran, most studies have primarily focused on clinical aspects such as treatment regimens, symptom severity, or short-term neonatal complications. Only limited research has explored the broader socioeconomic and epidemiologic determinants that influence neonatal

outcomes, particularly within the Iranian context where cultural, social, and health-system factors may differ substantially from other settings. Furthermore, few studies have employed a case-control design with a contemporaneous control group, which is essential for distinguishing the specific contribution of maternal substance use disorder from background socioeconomic influences. Addressing these gaps was the main rationale for conducting the present study.

Materials and methods

Study setting

This is a case-control study conducted on infants diagnosed with NOWS, who were hospitalized in the neonatal intensive care units, and a group of healthy babies born in the same hospital affiliated to Shiraz University of Medical Sciences, Iran between 2022 and 2023.

Study population

The study population comprises infants admitted to the Neonatal Intensive Care Unit (NICU) diagnosed with withdrawal or deprivation syndrome from opioid exposure and healthy neonates born at the same hospital. Cases were identified through a complete census, including all infants meeting the inclusion criteria of withdrawal diagnosis. Exclusion criteria encompassed infants with major birth defects, significant respiratory support beyond 72 h, neonatal seizure disorders, those requiring major surgical interventions, and instances of incomplete records/questionnaires.

Control subjects were selected based on matching criteria, including infant gender and maternal age, to ensure compatibility with the case group. Controls were drawn from infants born in the same hospital, with pregnancies and post-birth periods devoid of complications. This stringent selection process aims to establish a well-defined and comparable study population for both cases (exposed) and controls (non-exposed).

Methods

The study variables include gestational age, birth weight, head circumference, sex, delivery method, Apgar score at 1 and 5 min, type of opioid exposure, infant nutrition (breast milk or formula), Finnegan score (mean and maximum), type of treatment received, length of hospitalization, discharge to home or childcare facilities, maternal age, duration of substance use disorder, type of substance use disorder (inhalation, oral, injection), parental education level, parental occupation and family economic status. The family's economic status is classified into three categories, below average, average, and above average, based on each mother's estimate of her family life. Socioeconomic status was evaluated primarily by maternal self-report regarding perceived family economic level

(below average, average, or above average). To enhance accuracy, self-reported information was cross-validated using objective indicators, including house ownership (owner, tenant, or homeless), size of the residence (in square meters), and residential area/neighborhood. These indicators were recorded by the interviewer during face-to-face data collection and provided additional context to support the validity of the socioeconomic data. Moreover, to better evaluate the economic situation, the ownership status (owner, tenant, homeless), neighborhood and residential area are also inquired in the questionnaire. Information related to infants and mothers is obtained prospectively through a questionnaire (supplementary file 1) completed by one of the nursing colleagues who has received full training in the field of project implementation, and completing the questionnaire through a face-to-face interview with the mother.

The Finnegan Neonatal Abstinence Scoring Tool (FNAST) is widely used to assess the severity of neonatal withdrawal and guide the initiation, adjustment, and weaning of pharmacologic therapy [19]. Comprising 21 comprehensive items, this tool evaluates diverse symptoms observed in infants exposed to opiates, including high-pitched crying, sleep patterns, hyperactive Moro reflex, tremors, muscle tone, sweating, yawning, nasal stuffiness, respiratory rate, and other withdrawal-related indicators. Each symptom is given a score of 1 to 3 based on severity, and a score of 5 for seizures. The minimum score is 0 and the maximum is 42. When the patient's score reaches above 8 on three consecutive occasions with an interval of 3 h or above 12 on two consecutive occasions with an interval of 3 h, oral morphine is started for the patient [20]. FANST was applied at 3-hour intervals throughout hospitalization, starting from the time of admission. A trained physician performed all scoring to ensure consistency.

Statistical analysis

First, we performed descriptive statistical analysis to summarize and describe the study data. Frequencies and percentages were used to describe categorical variables while means and standard deviations were employed to describe continuous variables. Independent sample t-test, ANOVA, and chi-square test were also used to assess the potential association between the mothers' socio-demographic background and their clinical status (non-exposed). Furthermore, to explore the risk indicators of the continuous variables (length of hospital stay for the case group) the Negative binomial regression model was fitted on the predictor variables such as age, sex, type of delivery, feeding, education level of parents, occupational status of parents, and economic level. In the first step, all the important predictors were considered for the bivariate analysis. Secondly, insignificant covariates

were removed until only predictors that showed acceptable association ($p < 0.2$) in the bivariate analysis were candidates to include in the final multivariate regression models. The incidence rate ratio (IRR) for each predictor was also estimated. It should be noted that the regression analysis of exploring the risk indicators for the length of hospital stay was only conducted for the case group, as the control group had stays of only one or two days. The data was analyzed using SPSS software version 26.0 and the statistical significance level for all tests was set to be 0.05.

Ethical consideration

All methods were performed following the relevant guidelines and regulations. Informed consent was taken from parents. This article is extracted from the MPH thesis by Dr. Roya Oboodi at Shiraz University of Medical Sciences the Medical Ethics Committee approved the study protocol of Shiraz University of Medical Sciences (Ethics code: IR.SUMS.MED.REC.1402.443).

Results

During the study period, there were 6,617 births, and 95 mothers were diagnosed with substance use disorder. This results in an incidence rate of 1.4%. A total of 217 neonates were enrolled, with 95 (43.7%) in the case group and 122 (56.2%) in the control group. The mean gestational age was 37.49 ± 1.76 weeks, and the mean birth weight was 2863.78 ± 501.41 g. The average maternal age was 32.12 ± 6.52 years. Demographic characteristics of the case and control patients are outlined in Tables 1 and 2.

The prevalence of substance use disorder varied among cases, with methadone being the most common substance at 45 (47.4%), followed by heroin (35, 36.8%), opium (26, 26.4%), other substances including crystal methamphetamine, hashish, marijuana, and tramadol. Type of substance abuse predominantly involved inhalation (62, 65.3%), followed by oral ingestion (47, 49.5%), and intravenous injection (1, 1.1%).

Among mothers in the case group, 38 (40%) reported that they had attempted to discontinue substance use during pregnancy, although most were unable to maintain abstinence until delivery.

Among the cases, 56 (58.9%) required medical treatment for NOWS, with 53 (55.8%) receiving morphine and 13 (13.7%) receiving phenobarbital. Additionally, 11 (11.6%) received both phenobarbital and morphine.

In terms of discharge disposition, 55 (57.9%) neonates from the case group were discharged home, 38 (40%) were transferred to social services, and 2 (2.1%) were discharged to prison.

A comparison between cases that receive medication and those that do not receive is shown in Table 3.

Table 1 The demographic and clinical characteristics of case and control patients

	Cases (exposed) N = 95			p-value	Control (non-exposed) N = 122	p-value
	Total	Medical therapy N = 56	Non-medical therapy N = 39			
Sex				0.5		0.84
• Male	40 (42.1%)	22 (39.2%)	18 (46.1%)		53 (43.4%)	
• Female	55 (57.9%)	34 (60.7%)	21 (53.8%)		69 (56.5%)	
Type of delivery				0.73		0.49
• NVD	37(38.9%)	21 (37.5%)	16 (41%)		42 (34.4%)	
• Cesarean section	58 (61.1%)	35 (65.5%)	23 (58.9%)		80 (65.5%)	
Nutrition type				0.69		< 0.001
• Breast milk	6 (6.3%)	3 (5.3%)	3 (7.6%)		122 (100%)	
• Formula	89 (93.7%)	53 (94.6%)	36 (92.3%)		0	
Mother's education degree				0.21		0.005
• illiterate	10 (10.5%)	4 (7.1%)	6 (15.3%)		23 (18.8%)	
• primary and high school	52 (54.7%)	31 (55.3%)	21 (53.8%)		40 (32.7%)	
• diploma	26 (27.4%)	14 (25%)	12 (30.7%)		38 (31.1%)	
• associated degree and more	7 (7.4%)	7 (12.5%)	0		21 (17.2%)	
Father's education degree				0.098		0.034
• illiterate	25 (26.3%)		15 (38.4%)		15 (12.2%)	
• primary and high school	42 (44.2%)		15 (38.4%)		55 (45%)	
• diploma	19 (20%)		6 (15.3%)		39 (31.9%)	
• associated degree and more	9 (9.5%)		3 (7.6%)		13 (10.6%)	
Mother's job				0.011		< 0.001
• House worker	65 (68.4%)		21 (53.8%)		117 (95.9%)	
• Others	30 (31.6%)		18		5 (4%)	
Father's job				0.19		< 0.001
• Unemployed	11 (11.6%)		7 (17.9%)		0	
• Hand worker	72 (75.8%)		26 (66.6%)		113 (92.6%)	
• Others	12 (12.6%)		6 (15.2%)		9 (7.3%)	
Marital status				0.43		< 0.001
• Married	71 (74.2%)		30 (76.9%)		122 (100%)	
• unmarried	24(25.8%)		9 (23.1%)		0	
Economic level				0.88		0.8
• Low	60 (63.2%)		26 (66.6%)		79 (64.7%)	
• Medium	30 (31.6%)		11 (28.2%)		42 (34.4%)	
• High	5 (5.3%)		2 (5.1%)		1 (0.8%)	
House status				0.16		< 0.001
• Owner	20 (21.1%)		6 (15.3%)		53 (43.4%)	
• Tenant	68 (71.6%)		28 (71.7%)		69 (56.5%)	
• Homeless	7 (7.4%)		5 (12.8%)		0	

Table 2 Independent samples test: (between case and control)

	Case (mean ± SD)	Control (mean ± SD)	P-Value
Age at Discharge	12.52 ± 7.23	1.22 ± 0.423	< 0.001
Gestational Age	36.93 ± 1.98	37.92 ± 1.43	< 0.001
Birth Weight	2661 ± 472.34	3021.68 ± 466.97	< 0.001
Head Circumference	33.28 ± 1.54	34.43 ± 1.25	< 0.001
Hospitalization Days	14.35 ± 7.42	1.22 ± 0.423	< 0.001
Mother Age	33.34 ± 6.25	31.18 ± 6.60	0.015
1 st min apgar	7.87 ± 1.63	8.39 ± 0.74	0.005
5th min apgar	9.06 ± 1.10	9.44 ± 0.58	0.001
House Meter	82.26 ± 66.02	121.44 ± 80.96	< 0.001

The analysis demonstrated a significant relationship between birth weight and age at discharge (p-value 0.049). Additionally, a significant association was observed between the maximum Finnegan score and hospitalization days (p-value 0.028).

The association between the type of drug abused and the need for treatment revealed a significant association with opium (p-value 0.029).

Moreover, maternal education beyond a diploma and the duration of hospitalization showed a significant association (p-value 0.03), confirmed by post hoc tests (p-value 0.039). House workers demonstrated a

Table 3 Characteristics of cases received medical or non-medical treatment

	Medical therapy (56)	Non-medical therapy (39)	p- value
Gestational age	36.98 ± 1.97	36.87 ± 2.02	0.79
Age at discharge	14.5 ± 7.42	9.6 ± 5.98	0.001
Apgar 1 st min.	7.98 ± 1.63	7.7 ± 1.63	0.44
Apgar 5th min.	9.2 ± 0.94	8.84 ± 1.28	0.11
Birth weight	2657.5 ± 551.34	2666 ± 334.49	0.93
Head circumference	33.21 ± 1.69	33.39 ± 1.32	0.57
Finnegan score	6.85 ± 1.85	3.96 ± 1.82	< 0.001
Maximum Finnegan score	12 ± 2.94	6.05 ± 2.23	< 0.001
Mother age	32.98 ± 6.31	33.87 ± 6.19	0.49
Duration of substance use disorder (years)	5.25 ± 4.8	4.5 ± 4.5	0.46
House meter	89.39 ± 65.33	72.02 ± 66.5	0.2

Table 4 Relationships between length of hospital stay and selected predictors (Negative binomial regression) (n = 90)

	Incident rate ratio	95% CI	p
Treatment			
Medical therapy	1.46	1.19–1.81	< 0.0001
Non-Medical			
Mothers Education			
Academic	1.54	1.06–2.25	< 0.0001
Non-academic			
Apgar 5 min	0.90		0.03
Duration of substance use disorder	0.95	0.88–1.04	0.08
Birth weight	0.99	0.98–1.01	0.06

95% CI = 95% Confidence Interval

significantly higher maximum Finnegan score compared to other occupations (p -value < 0.001). Neonates of fathers with employment were more likely to be discharged home in parental custody than those without employment or engaged in criminal activities (p -value 0.014).

Marital status did not significantly impact birth indices or the need for medical treatment but was associated with discharge disposition (p -value < 0.001). Discharge to social services was more common among tenants or homeless individuals (p -value 0.007).

Notably, neonates discharged to community (social) services experienced a longer length of hospital stay compared to those discharged home (p -value < 0.001), although they had lower maximum Finnegan scores (p -value 0.022).

Negative binomial regressions

To determine how the mothers' socio-demographic background and delivery history factors affect the length of hospital stay for the case group, all the mentioned factors

were analyzed in bivariate analysis, treatment status, mother's education, years of a substance use disorder, birth weight and Apgar score in five minutes showed a p -value of less than 0.2 and were considered for inclusion in the regression model. The final regression model (Table 4) revealed that the length of hospital stays for medical subjects was 1.46 times longer compared to non-medical subjects who did not seek treatment (IRR = 1.46, p < 0.0001). Academic mothers were expected to have a 1.54 times greater rate of hospital stay compared to nonacademic mothers (IRR = 1.54, p < 0.0001). Additionally, the rate ratio for the Apgar score at 5 min was 0.9, indicating that infants with a lower score had a higher risk of prolonged hospital stay (IRR = 0.9, p < 0.03). Other predictors did not show a significant effect in the final regression model.

Discussion

The findings of this study reveal several critical insights into maternal substance abuse and its impact on neonatal health. Methadone emerged as the most frequently used substance among the study participants. About 60% of neonates in the case group were affected by NOWS, necessitating treatment with morphine and phenobarbital. Additionally, a considerable number of neonates in the case group (40%) were transferred to social services. The analyses further identified significant associations between maternal education, occupation, paternal employment status, and discharge disposition. These results underscore the complex interplay between maternal substance abuse, treatment interventions, and neonatal outcomes, highlighting the need for comprehensive care strategies for this vulnerable population.

In our study, we observed that 40% of women made attempts to discontinue substance use during pregnancy. This finding aligns with trends noted in other studies, suggesting a propensity among pregnant women to cease drug usage, resulting in positive outcomes such as the delivery of healthy babies [21, 22]. However, it's essential to note that despite these efforts, a significant portion of pregnant women with substance use disorders may still face challenges in completely abstaining from drugs throughout their pregnancies. This underscores the complex nature of substance use disorder and the need for comprehensive support systems for expectant mothers struggling with substance use.

In the present study, marital status emerged as a significant factor influencing the discharge destination of neonates. Among the mothers included in the case group, 74.2% were married, contrasting with the control groups where all mothers were married. This finding emphasizes the potential association between marital status and the likelihood of infant discharge home. According to this claim, only 57.9% of the infants in the case group and

100% of the infants in the control group were discharged home. An investigation spanning from 2010 to 2017 in Arizona unveiled that a significant proportion of women dependent on opioids during childbirth were unmarried [23]. In a study conducted in Tennessee, it was observed that mothers of neonates with NOWS were more likely to be unmarried compared to those without NOWS (72.7% vs. 58.4%, $p < 0.0001$) [24]. In a research project undertaken in Alaska hospitals focusing on neonates afflicted with neonatal opioid withdrawal syndrome, findings revealed that 70% of these infants were discharged to their parents, while the remaining were placed with relatives or foster caregivers. Notably, mothers who used methamphetamine and stimulants exhibited a reduced likelihood of having their neonates discharged to their homes. Furthermore, the presence of parents during the hospitalization period showed a significant association between neonatal discharge to parental care at home [16] which is similar to our results. Historical research has indicated the positive outcomes associated with integrating medications for opioid use disorder alongside parenting and child welfare support, facilitating parents' ability to retain custody of their children [25].

In this investigation, the majority of the case and control groups had a low economic level. Therefore, the lack of difference in the economic status in the case and control groups and its possible effects on the consequences of neonatal abstinence syndrome can be justified. In a comprehensive study encompassing 580 counties across the United States, researchers examined the relationship between country-level economic indicators and neonatal abstinence syndrome. The findings indicated that elevated long-term unemployment rates were linked with increased occurrences of NOWS, implying that both historical and contemporary economic factors play a role in influencing NOWS rates [26].

Vesoulis and colleagues addressed a knowledge gap by retrospectively analyzing 129 full-term infants diagnosed with NOWS. The researchers discovered a significant positive association between the level of deprivation and prolonged hospitalization for NOWS ($r = 0.55$, $P < 0.01$), which persisted even after adjusting for factors like pharmacological treatment, infant custody, race, and gestational age [15].

We observed that neonates whose mothers had higher levels of academic education tended to have longer hospital stays. This finding contrasts with the results of a study conducted in Florida regarding the length of stay for patients with NOWS revealed that neonates born to mothers with a high school or less had a length of stay of 12.4 days, whereas neonates born to mothers with greater than a high school education had a slightly shorter length of stay of 11.7 days (p -value = 0.032) [27]. The association between higher maternal education and prolonged

neonatal length of stay is unexpected and warrants careful interpretation. Several hypotheses may explain this paradox. One possibility is that local administrative or social-welfare procedures, such as discharge clearance or placement planning, may indirectly prolong hospitalization in some subgroups, though our data cannot confirm this. Another hypothesis is that mothers with higher education may be more likely to engage with health services or accept pharmacologic treatment for NOWS, both of which are associated with longer inpatient stays in prior research. Alternatively, unmeasured confounding variables — including maternal comorbidities, type of opioid exposure, or custody-related factors — may account for the observed association. Consistent with our findings, a recent study in the United States also reported significant variability in neonatal length of stay that was influenced not only by maternal demographics but also by hospital-level and community-level factors, underscoring the complex, context-dependent nature of this outcome [27]. In Hosseini et al.'s study, it was found that mothers of neonates with NOWs had significantly lower levels of education compared to mothers of healthy neonates. Additionally, the economic status of families in the NOWs group was also lower [28].

Parents with substance use disorder frequently face financial instability and uncertainty regarding access to food and housing. They may also experience a disorderly home environment, irregular employment patterns, domestic conflicts, social stigma or isolation, periods of incarceration, and heightened stress [29, 30]. Despite initial maternal intentions and supportive interventions, 27% of offspring born to mothers with substantial substance use disorders required the intervention of child protective services during their preschool years [31]. In our study, we observed that 42.1% of patients were discharged to locations other than their homes, such as social care facilities. This trend was particularly notable among individuals facing circumstances such as homelessness or tenancy, unmarried maternal status, and paternal unemployment. Despite exhibiting lower Finnegan scores, indicating less severe withdrawal symptoms, these individuals tended to experience prolonged hospital stays. This delay was primarily attributed to the administrative processes associated with transferring them to welfare or social care centers. The higher proportion of cases transferred to social care centers in our study could be attributed to cultural factors and societal stigmatization surrounding children born to unmarried women. Additionally, economic challenges faced by unmarried women and unemployed parents may exacerbate these circumstances, making it more difficult for them to provide adequate care for their children. These complex socio-cultural and economic dynamics likely

contribute to the decision to transfer children to social care centers in such cases.

In the case group, most of the babies were fed formula and lost the benefits of breastfeeding. In addition, these families have low economic backgrounds and face significant obstacles to preparing formula. Although methadone can be deemed safe during breastfeeding [32], in our study, the majority of mothers were subjected to abuse methadone, and they did not utilize it as a means to overcome their addiction. Guilt and misinformation about the safety of breastfeeding while undergoing medication-assisted treatment are barriers that prevent mothers from breastfeeding their children [33]. In our study, about 40% of neonates were discharged to health services and were deprived of maternal breastfeeding. Others were formula-fed due to unsafe opioid doses or family preference, despite being provided with information about breastfeeding.

The strength of this study lies in its novelty, particularly in its thorough examination of socioeconomic factors, which have been less comprehensively explored in previous research. Additionally, the inclusion of a control group significantly enhances the validity of the findings by enabling a comparison between cases and controls, thus facilitating more robust conclusions. This study has several limitations. First, it was conducted at a single center that predominantly serves a population from lower socioeconomic backgrounds, which may limit the generalizability of our findings. Second, socioeconomic data were primarily based on maternal self-report, although we attempted to cross-check this information with objective indicators such as house ownership, residence size, and neighborhood. Nevertheless, recall or reporting bias cannot be excluded. Third, as with most observational studies, unmeasured confounding factors may have influenced the observed associations, and causality cannot be firmly established. Future multi-center studies with larger and more diverse populations and more comprehensive socioeconomic measures are warranted to validate and expand upon these findings.

Conclusion

This study sheds light on the intricate relationship between maternal substance abuse, socioeconomic factors, and neonatal health outcomes. By identifying significant associations between maternal socioeconomic status and discharge disposition, the study underscores the need for targeted interventions to address the complex challenges faced by mothers with substance use disorder and their infants. Furthermore, the findings highlight the importance of comprehensive care strategies that take into account not only medical treatment but also socioeconomic support systems for this vulnerable population. Ultimately, understanding and addressing

these factors can have far-reaching implications for society, including improved neonatal health outcomes, reduced burden on healthcare systems, and enhanced support for families affected by substance abuse. This work contributes valuable insights that can inform policy-making, healthcare delivery, and community interventions aimed at mitigating the impact of maternal substance abuse on neonatal health and social well-being.

Abbreviations

NAS	Neonatal Abstinence Syndrome
NOWS	Neonatal Opioid Withdrawal Syndrome
NICU	Neonatal Intensive Care Unit
FNAST	Finnegan Neonatal Abstinence Scoring Tool
SPSS	Statistical Package for the Social Sciences
MPH	Master of Public Health
IRR	Incidence Rate Ratio
CI	Confidence Interval
US	United States

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12884-025-08377-3>.

Supplementary Material 1.

Authors' contributions

RO Concept and design, writing the draft, editing the manuscript. HB Concept and design, writing the draft, editing the manuscript. MDF analyzes the data, writes the draft, and edits the manuscript. RB Concept and design, writing the draft, editing the manuscript. SZ acquisition of data, writing the draft. AS Concept and design, writing the draft, editing the manuscript, supervising.

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Data availability

Materials and data provided in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All methods were performed in accordance with the relevant guidelines and regulations. Informed consent was taken from parents. This article is extracted from the MPH thesis by Dr. Roya Oboudi at Shiraz University of Medical Sciences the Medical Ethics Committee approved the study protocol of Shiraz University of Medical Sciences (Ethics code: IR.SUMS.MED.REC.1402.443).

Competing interests

The authors declare no competing interests.

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